

Обработка дублей для задания 2.2

```
--Проверка дубликатов в deal_info
select deal_rk, effective_from_date, COUNT(*)
from rd.deal_info
group by deal_rk, effective_from_date
HAVING COUNT(*) > 1;
```

	123 deal_rk	effective_from_date	123 count
1	4 531 242	2023-08-11	2
2	2 594 431	2023-03-15	2

```
--Проверка дубликатов в loan_holiday
select deal_rk, effective_from_date, COUNT(*)
from rd.loan_holiday
group by deal_rk, effective_from_date
HAVING COUNT(*) > 1;
```

	123 deal_rk	effective_from_date	123 count
1	4 531 242	2023-08-11	2
2	2 594 431	2023-03-15	2

```
--Проверка дубликатов в product
select product_rk, effective_from_date, COUNT(*)
from rd.product
group by product_rk, effective_from_date
HAVING COUNT(*) > 1;
```

	123 product_rk	effective_from_date	123 count
1	1 607 713	2023-01-01	2
2	1 842 604	2023-08-11	2
3	1 667 353	2023-01-01	2
4	1 303 986	2023-08-11	2
5	1 570 019	2023-01-01	2
6	1 292 075	2023-08-11	2
7	1 668 282	2023-03-15	2
8	1 169 710	2023-08-11	2
9	1 362 699	2023-08-11	2
10	1 956 251	2023-03-15	2
11	1 385 244	2023-08-11	2
12	1 855 120	2023-08-11	2
13	1 168 289	2023-08-11	2
14	1 969 939	2023-08-11	2
15	1 768 848	2023-01-01	2
16	1 583 303	2023-08-11	2
17	1 979 096	2023-03-15	2
18	1 308 366	2023-03-15	2

С дубликатами витрина выглядит следующим образом:

```
select effective_from_date, COUNT(*)
from dm.loan_holiday_info
group by effective_from_date
order by effective_from_date;
```

effective_from_date	count
2023-01-01	3 008
2023-03-15	3 510
2023-08-11	3 522

Вариант 1:

```
CREATE OR REPLACE PROCEDURE dm.refresh_loan_holiday_info()
LANGUAGE plpgsql
as $$
declare
    v_start_time TIMESTAMP;
    v_end_time TIMESTAMP;
    v_duration_sec INTEGER;
    v_rows INTEGER;
    v_error_message TEXT;
    v_process_name VARCHAR(255) := 'refresh_loan_holiday_info';
    v_source_table VARCHAR(255) := 'dm.loan_holiday_info';
BEGIN
    -- Запоминаем время начала
    v_start_time := NOW();

    -- Логируем начало выполнения
    insert into "LOGS"."ETL_LOG" (process_name, cur_date, start_time, status, source,
user_name)
        values (v_process_name, CURRENT_DATE, v_start_time, 'RUNNING', v_source_table,
CURRENT_USER);

    -- Очищаем витрину
    truncate table dm.loan_holiday_info;

    -- Заполняем витрину по прототипу
    insert into dm.loan_holiday_info (
        deal_rk, effective_from_date, effective_to_date, agreement_rk,
client_rk, department_rk, product_rk,
        product_name, deal_type_cd, deal_start_date, deal_name, deal_number,
deal_sum, loan_holiday_type_cd, loan_holiday_start_date,
        loan_holiday_finish_date, loan_holiday_fact_finish_date,
loan_holiday_finish_flg, loan_holiday_last_possible_date)
        with deal_raw as (
            select
                deal_rk, deal_num, deal_name, deal_sum, client_rk, agreement_rk,
deal_start_date,
                department_rk, product_rk, deal_type_cd, effective_from_date,
effective_to_date,
                COUNT(*) OVER (PARTITION BY deal_rk, effective_from_date) as dup_count
            from rd.deal_info
        ),
        deal as (
```

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        select * from deal_raw
        where dup_count = 1
    ),
    loan_holiday_raw as (
        select
            deal_rk, loan_holiday_type_cd, loan_holiday_start_date,
loan_holiday_finish_date,
            loan_holiday_fact_finish_date, loan_holiday_finish_flg,
loan_holiday_last_possible_date,
            effective_from_date, effective_to_date,
            COUNT(*) OVER (PARTITION BY deal_rk, effective_from_date) as dup_count
        from rd.loan_holiday
    ),
    loan_holiday as (
        select * from loan_holiday_raw
        where dup_count = 1
    ),
    product_raw as (
        select
            product_rk, product_name, effective_from_date, effective_to_date,
            COUNT(*) OVER (PARTITION BY product_rk, effective_from_date) as dup_count
        from rd.product
    ),
    product as (
        select product_rk, product_name, effective_from_date, effective_to_date
        from product_raw
        where dup_count = 1
    ),
    holiday_info as (
        select
            d.deal_rk
            ,lh.effective_from_date
            ,lh.effective_to_date
            ,d.deal_num as deal_number --Номер сделки
            ,lh.loan_holiday_type_cd --Ссылка на тип кредитных каникул
            ,lh.loan_holiday_start_date --Дата начала кредитных каникул
            ,lh.loan_holiday_finish_date --Дата окончания кредитных каникул
            ,lh.loan_holiday_fact_finish_date --Дата окончания кредитных каникул
фактическая
            ,lh.loan_holiday_finish_flg --Признак прекращения кредитных каникул по
инициативе заёмщика
            ,lh.loan_holiday_last_possible_date --Последняя возможная дата кредитных
каникул
            ,d.deal_name --Наименование сделки
            ,d.deal_sum --Сумма сделки
            ,d.client_rk --Ссылка на контрагента
            ,d.agreement_rk --Ссылка на договор
            ,d.deal_start_date --Дата начала действия сделки
            ,d.department_rk --Ссылка на ГО/филиал
            ,d.product_rk -- Ссылка на продукт
            ,p.product_name -- Наименование продукта
            ,d.deal_type_cd -- Наименование типа сделки
        from deal d
        left join loan_holiday lh on 1=1
            and d.deal_rk = lh.deal_rk
            and d.effective_from_date = lh.effective_from_date
        left join product p on p.product_rk = d.product_rk
            and p.effective_from_date = d.effective_from_date
    )
select
    deal_rk,
    effective_from_date,

```

```

effective_to_date,
agreement_rk,
client_rk,
department_rk,
product_rk,
product_name,
deal_type_cd,
deal_start_date,
deal_name,
deal_number,
deal_sum,
loan_holiday_type_cd,
loan_holiday_start_date,
loan_holiday_finish_date,
loan_holiday_fact_finish_date,
loan_holiday_finish_flg,
loan_holiday_last_possible_date
from holiday_info;

-- Получаем количество вставленных строк
GET DIAGNOSTICS v_rows = ROW_COUNT;

-- Завершаем логирование
v_end_time := NOW();
v_duration_sec := EXTRACT(EPOCH FROM (v_end_time - v_start_time));

-- Логлируем успешное завершение
insert into "LOGS"."ETL_LOG" (process_name, cur_date, start_time, end_time,
duration_sec, status, rows_written, source, user_name)
values (v_process_name, CURRENT_DATE, v_start_time, v_end_time,
v_duration_sec, 'SUCCESS', v_rows, v_source_table, CURRENT_USER);

raise notice 'Витрина dm.loan_holiday_info перезагружена. Добавлено строк: %',
v_rows;

EXCEPTION
WHEN OTHERS THEN
    GET STACKED DIAGNOSTICS v_error_message = MESSAGE_TEXT;
    v_end_time := CURRENT_TIMESTAMP;
    v_duration_sec := EXTRACT(EPOCH FROM (v_end_time - v_start_time));

    insert into "LOGS"."ETL_LOG" (process_name, cur_date, start_time, end_time,
duration_sec, status, error_message, rows_error, source, user_name)
values (v_process_name, CURRENT_DATE, v_start_time, v_end_time,
v_duration_sec, 'ERROR', v_error_message, 1, v_source_table, CURRENT_USER);

    raise notice 'Ошибка при перезагрузке витрины dm.loan_holiday_info: %',
v_error_message;
END;
$$;

```

То есть на этапе выполнения процедуры будем «помечать» все записи невалидными и исключать их из расчета (любая запись, для которой существует более одного варианта на ключ).

Витрина примет следующий вид:

	effective_from_date	123 count
1	2023-01-01	3 000
2	2023-03-15	3 498
3	2023-08-11	3 498

2023-01-01: дубликаты только в product (8 конфликтующих строк), они исключены

2023-03-15: дубликаты в deal_info (2 строки), loan_holiday (2 строки), product (8 строк). При JOIN теряются обе конфликтующие сделки - 3498

2023-08-11: аналогично, теряются 2 сделки - 3498

Вариант 2:

```
CREATE OR REPLACE PROCEDURE dm.refresh_loan_holiday_info()
LANGUAGE plpgsql
as $$
declare
    v_start_time TIMESTAMP;
    v_end_time TIMESTAMP;
    v_duration_sec INTEGER;
    v_rows INTEGER;
    v_error_message TEXT;
    v_process_name VARCHAR(255) := 'refresh_loan_holiday_info';
    v_source_table VARCHAR(255) := 'dm.loan_holiday_info';
BEGIN
    -- Запоминаем время начала
    v_start_time := NOW();

    -- Логируем начало выполнения
    insert into "LOGS"."ETL_LOG" (process_name, cur_date, start_time, status, source,
user_name)
        values (v_process_name, CURRENT_DATE, v_start_time, 'RUNNING', v_source_table,
CURRENT_USER);

    -- Очищаем витрину
    truncate table dm.loan_holiday_info;

    -- Заполняем витрину по прототипу
    insert into dm.loan_holiday_info (
        deal_rk, effective_from_date, effective_to_date, agreement_rk,
client_rk, department_rk, product_rk,
        product_name, deal_type_cd, deal_start_date, deal_name, deal_number,
deal_sum, loan_holiday_type_cd, loan_holiday_start_date,
        loan_holiday_finish_date, loan_holiday_fact_finish_date,
loan_holiday_finish_flg, loan_holiday_last_possible_date)
        with deal_raw as (
            select
                deal_rk, deal_num, deal_name, deal_sum, client_rk, agreement_rk,
deal_start_date,
                department_rk, product_rk, deal_type_cd, effective_from_date,
effective_to_date,
                ROW_NUMBER() OVER (PARTITION BY deal_rk, effective_from_date ORDER BY
deal_sum) as rn
            from rd.deal_info
        ),
        deal as (
```

```

        select * from deal_raw
        where rn = 1
    ),
    loan_holiday_raw as (
        select
            deal_rk, loan_holiday_type_cd, loan_holiday_start_date,
            loan_holiday_finish_date,
            loan_holiday_fact_finish_date, loan_holiday_finish_flg,
            loan_holiday_last_possible_date,
            effective_from_date, effective_to_date,
            ROW_NUMBER() OVER (PARTITION BY deal_rk, effective_from_date ORDER BY
            loan_holiday_last_possible_date) as rn
        from rd.loan_holiday
    ),
    loan_holiday as (
        select * from loan_holiday_raw
        where rn = 1
    ),
    product_raw as (
        select
            product_rk, product_name, effective_from_date, effective_to_date,
            ROW_NUMBER() OVER (PARTITION BY product_rk, effective_from_date ORDER BY
            product_name) as rn
        from rd.product
    ),
    product as (
        select product_rk, product_name, effective_from_date, effective_to_date
        from product_raw
        where rn = 1
    ),
    holiday_info as (
        select
            d.deal_rk
            ,lh.effective_from_date
            ,lh.effective_to_date
            ,d.deal_num as deal_number --Номер сделки
            ,lh.loan_holiday_type_cd --Ссылка на тип кредитных каникул
            ,lh.loan_holiday_start_date --Дата начала кредитных каникул
            ,lh.loan_holiday_finish_date --Дата окончания кредитных каникул
            ,lh.loan_holiday_fact_finish_date --Дата окончания кредитных каникул
            ,lh.loan_holiday_finish_flg --Признак прекращения кредитных каникул по
            инициативе заёмщика
            ,lh.loan_holiday_last_possible_date --Последняя возможная дата кредитных
            каникул
            ,d.deal_name --Наименование сделки
            ,d.deal_sum --Сумма сделки
            ,d.client_rk --Ссылка на контрагента
            ,d.agreement_rk --Ссылка на договор
            ,d.deal_start_date --Дата начала действия сделки
            ,d.department_rk --Ссылка на ГО/филиал
            ,d.product_rk -- Ссылка на продукт
            ,p.product_name -- Наименование продукта
            ,d.deal_type_cd -- Наименование типа сделки
        from deal d
        left join loan_holiday lh on 1=1
            and d.deal_rk = lh.deal_rk
            and d.effective_from_date = lh.effective_from_date
        left join product p on p.product_rk = d.product_rk
            and p.effective_from_date = d.effective_from_date
    )
select

```

```

deal_rk,
effective_from_date,
effective_to_date,
agreement_rk,
client_rk,
department_rk,
product_rk,
product_name,
deal_type_cd,
deal_start_date,
deal_name,
deal_number,
deal_sum,
loan_holiday_type_cd,
loan_holiday_start_date,
loan_holiday_finish_date,
loan_holiday_fact_finish_date,
loan_holiday_finish_flg,
loan_holiday_last_possible_date
from holiday_info;

-- Получаем количество вставленных строк
GET DIAGNOSTICS v_rows = ROW_COUNT;

-- Завершаем логирование
v_end_time := NOW();
v_duration_sec := EXTRACT(EPOCH FROM (v_end_time - v_start_time));

-- Логируем успешное завершение
insert into "LOGS"."ETL_LOG" (process_name, cur_date, start_time, end_time,
duration_sec, status, rows_written, source, user_name)
values (v_process_name, CURRENT_DATE, v_start_time, v_end_time,
v_duration_sec, 'SUCCESS', v_rows, v_source_table, CURRENT_USER);

raise notice 'Витрина dm.loan_holiday_info перезагружена. Добавлено строк: %',
v_rows;

EXCEPTION
WHEN OTHERS THEN
GET STACKED DIAGNOSTICS v_error_message = MESSAGE_TEXT;
v_end_time := CURRENT_TIMESTAMP;
v_duration_sec := EXTRACT(EPOCH FROM (v_end_time - v_start_time));

insert into "LOGS"."ETL_LOG" (process_name, cur_date, start_time, end_time,
duration_sec, status, error_message, rows_error, source, user_name)
values (v_process_name, CURRENT_DATE, v_start_time, v_end_time,
v_duration_sec, 'ERROR', v_error_message, 1, v_source_table, CURRENT_USER);

raise notice 'Ошибка при перезагрузке витрины dm.loan_holiday_info: %',
v_error_message;
END;
$$;

```

Здесь для каждой группы дубликатов выбирается первая запись по заданному правилу сортировки, остальные исключаются.

Результаты витрины:

	effective_from_date	123 count
1	2023-01-01	3 000
2	2023-03-15	3 499
3	2023-08-11	3 499

2023-01-01: дубликаты в product устранены, сделки не затронуты

2023-03-15: из двух конфликтующих сделок остаётся одна, вторая теряется

2023-08-11: аналогично, остаётся одна сделка из двух